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Research article

Establishing "Glycoantigens" as a cross-field concept in Glycobiology, transfusion, transplant, and immunology☆

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ABSTRACT

Cell-surface glycans (carbohydrate structures) serve as critical antigens in transfusion and transplantation, yet their study remains fragmented across disciplines. The ABO blood group system, the most clinically tested glycan compatibility system, is still assessed using the same broad phenotypic classifications (A, B, AB, O) established over 125 years ago, which fail to capture the underlying structural diversity of glycans or account for tissue-specific expression patterns. Beyond ABO, numerous other glycoantigens remain almost entirely neglected in clinical compatibility testing, despite their potential relevance in antibody-mediated rejection, xeno-transplantation, and unexplained graft complications. I propose establishing "glycoantigens" as a unified cross-disciplinary concept encompassing immunogenic glycan epitopes relevant to compatibility in transfusion and transplantation. Modern tools, including next-generation and nanopore sequencing of glycosyltransferase genes, glycan microarrays, Luminex-based bead immunoassays, and advanced mass spectrometry, now enable systematic characterization of these complex structures. I outline a tiered clinical implementation pathway, propose an integrated glycoantigen database linking genomic, structural, expression, and serologic data, and describe a Glycoantigen Prediction Pipeline that translates multi-gene genotypes through predicted enzyme activities into tissue-specific glycoantigen profiles with antibody risk estimates. Recognizing glycoantigens as a distinct research area would foster collaboration between glycobiologists and transplant immunologists, enable therapeutic innovations, and potentially explain cases of graft rejection not attributable to HLA or other known factors. Although the initial focus is transfusion and transplantation, anti-glycan antibodies are clinically significant across allergy, neurology, gastroenterology, infectious disease, and oncology, and the proposed database could ultimately serve as a broader resource for glycoantigen biology in medicine.

1. Introduction: glycans as underappreciated antigens

In blood transfusion and organ transplantation, cell-surface glycans (carbohydrate structures) determine the ABO blood group — among the most immunogenic of all blood group antigens, necessitating strict donor-recipient matching to avoid catastrophic immune reactions [1,2]. However, current ABO typing methods provide only phenotypic, single-view representations, classifying individuals into the same broad A, B, AB, or O categories established over 125 years ago. Even refinements using lectins (distinguishing A_1 from non-A_1) fail to capture the full structural diversity underlying these broad phenotypes. Hamm et al. recently used glycan microarrays to show that lectins commonly used in clinical ABO testing have poorly characterized glycan specificities with significant off-target binding, whereas monoclonal antibodies demonstrated superior specificity [3], highlighting the importance of evaluating and adopting improved glycan-detection reagents as they become available. Each ABO "type" encompasses a heterogeneous collection of related glycan structures that vary in branching, carrier molecules (glycoproteins vs. glycolipids), and core chain types.

A striking example of how transplantation demands have exposed the limitations of conventional serologic methods comes from A_2-to-B kidney transplantation. Because group A_2 donors express less A antigen, their kidneys can be safely transplanted into group B recipients, a critical pathway for reducing the disproportionately long waitlist times faced by

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blood group B candidates. This clinical need drove A_1/A_2 subtyping using anti-A_1 lectin at a scale far exceeding anything required for routine transfusion. However, we showed in a multi-center study that *ABO* genotyping identified substantially more A_2 donors than lectin testing [4]. Because the A_2 subtype is defined by a single causative genetic change (c.1061delC) in the vast majority of cases, genotyping proved more reliable than serologic methods, which are confounded by biological variability, minor variations in glycan levels, and clinical factors such as prior transfusions. This experience illustrates a key theme of this paper: when transplantation raises the stakes of glycoantigen classification, the broad phenotypic tools inherited from transfusion medicine prove inadequate, and a cross-disciplinary approach integrating molecular genetics with glycobiology yields superior results.

Moreover, all standard blood group testing focuses exclusively on red blood cells and does not account for tissue-specific expression of the glycosyltransferase enzymes that synthesize these antigens. The same individual may express different glycan patterns on vascular endothelium, epithelial tissues, and solid organs compared to their circulating red cells. This tissue-specific variation has profound implications for transplantation: a donor-recipient pair deemed "ABO-incompatible" by standard RBC typing might in fact be compatible at the tissue glycoantigen level, a distinction that current testing cannot make. Because A-II glycans (A antigen on type 2 chains) are the sole A-antigen subtype expressed on kidney vascular endothelium, the relevant question for A_2-to-B and A_2-to-O transplantation is whether a candidate has antibodies against A-II specifically — yet conventional anti-A hemagglutination titers cannot make this distinction. Halpin et al. recently demonstrated the clinical significance of this gap: using a Luminex-based multiplex bead immunoassay that could distinguish anti-A-II from other anti-A specificities, they suggested that a precise understanding of glycoantigen-specific antibodies could identify transplant opportunities in candidates previously deemed incompatible by conventional broad-specificity testing [5]. Similarly, research using ABH-glycan microarrays has shown that pediatric recipients of ABO-incompatible heart grafts did not develop detectable antibodies against the specific A/B glycoantigen subtypes present in the donor heart, evidence of glycan-subtype-specific B-cell tolerance that would be entirely missed by conventional hemagglutination assays [6].

Beyond kidney transplantation, ABO expression patterns differ markedly across solid organs relevant to transplantation. In the liver, A and B antigens are expressed predominantly on biliary epithelium and vascular endothelium rather than hepatocytes, which may be one biological factor contributing to the relative feasibility of ABO-incompatible liver transplantation compared with ABO-incompatible kidney transplantation. In the heart, ABH antigens are expressed on vascular endothelium and reported on epicardial/mesothelial surfaces, but not on cardiac myocytes; in the lung, ABH-related antigens are expressed by respiratory epithelium and vascular endothelium, with patterns varying by cell type, glycan precursor, and secretor status. The success of ABO-incompatible heart and lung transplantation in infants is attributable chiefly to the immature isohemagglutinin response in early life rather than to absence of tissue antigen expression [1,7]. These organ-specific patterns are not captured by red-cell phenotyping and, like the A-II example above, illustrate why a glycoantigen framework that integrates tissue-level expression with genotype and antibody data is needed to support transplantation decisions.

Beyond human allotransplantation, glycoantigens other than ABO are almost entirely neglected in routine compatibility testing, despite evidence that they could play a role in graft outcomes. Lewis blood group antibodies, for example, have been reported to cause antibody-mediated rejection in kidney transplant recipients [8], and other glycan blood group systems could have similar unrecognized significance. Meanwhile, the study of glycans remains siloed: glycobiologists, transfusion specialists, and xenotransplantation researchers work with different frameworks and terminology, and no unifying concept bridges detailed glycan science with clinical compatibility testing.

Although the term "glycoantigen" appears sporadically in the scientific literature (in oncology, where carbohydrate antigens have been noted as neglected [9], and in xenotransplantation and vaccine research), it has not been defined or adopted as a unified framework, and those prior uses would naturally fall within the broader concept I propose here. To address this gap, I propose the term "glycoantigens" as a cross-disciplinary category for glycan structures that serve as immunologically significant antigens in transfusion and transplantation.

1.1. Beyond ABO: a spectrum of human glycoantigens

ABO is the tip of the iceberg. Each glycoantigen system is defined by a specific enzyme polymorphism: if a person lacks the enzyme that adds a particular sugar, they will not express that glycan epitope and might form antibodies if exposed [1]. The sequential, interconnected nature of glycan biosynthesis means that a deficiency in one enzyme can have cascading effects on multiple downstream antigens, creating enormous combinatorial diversity from a relatively limited set of glycosyltransferases. Multiple other recognized human blood group systems are defined by immunogenic glycan structures [1], several of which have direct relevance to transfusion and transplantation.

The H antigen (*FUT1/FUT2*) serves as the biosynthetic precursor upon which A and B antigens are built. The Lewis system (*FUT3*) has traditionally been considered clinically insignificant for transfusion, yet Lewis antigens are expressed on kidney tubular cells, and studies have documented antibody-mediated renal allograft rejection in Lewis-incompatible recipients with anti-Lewis antibodies [8]. Lewis antigen expression primarily depends on the combined activities of *FUT3*, which encodes the Lewis α1,3/4-fucosyltransferase, and *FUT2*, which determines secretor status by generating type 1H structures that can be further modified to form Le^b. Lewis and related antigens are widely expressed beyond red cells, including gastrointestinal epithelium and vascular endothelium, supporting their potential relevance in solid organ transplantation. Mechanistically, Lewis antigens are glycolipids synthesized in secretory tissues and passively adsorbed onto vascular endothelium and circulating red cells from plasma, so a donor's Lewis phenotype can present a target for preformed recipient anti-Lewis antibodies at the graft endothelium. Because Lewis and secretor status (*FUT3* and *FUT2*) is not part of standard donor-recipient matching, a Lewis-related contribution to antibody-mediated rejection can go unrecognized. Incorporating Lewis phenotype into a glycoantigen framework would allow anti-Lewis antibody screening to be considered prospectively in sensitized candidates, or in cases of otherwise unexplained rejection, rather than only after graft injury. More generally, minor glycoantigens are clinically relevant in transplantation precisely because they remain invisible to current testing, so any contribution they make to graft outcomes cannot presently be assessed. Thus, interpretation of Lewis-related glycan expression requires integration of multiple enzymatic pathways, including *FUT3*-mediated Lewis fucosylation, *FUT2*-mediated secretor/H antigen synthesis. The I/i system (*GCNT2*) is primarily associated with cold agglutinin disease and autoimmune hemolysis. The Sd^a blood group system (*B4GALNT2*) has gained renewed prominence through xenotransplantation research, as the Sd^a antigen and related glycan structures are one of three major glycan targets in pig-to-human incompatibility.

Understanding the interrelated nature of these glycoantigens is essential. Each antigen can exist on multiple core chain types, creating structural variants that share the same terminal epitope but differ in their underlying backbone and tissue distribution [10]. Two fucosyltransferases (*FUT1* and *FUT2*) synthesize H antigens on different precursor chains, producing at least four structural forms (H types 1–4), with distinct forms predominating in secretory epithelia versus erythrocytes and vascular endothelium. Glycoantigens can also combine to create complex epitopes, such as ALe^b and BLe^b, where Lewis and ABO modifications co-exist on the same chain [10]. This structural diversity means that the same nominal blood type can present different glycan

profiles in different tissues — a distinction invisible to conventional serologic testing but potentially significant for transplant compatibility.

These established blood group systems represent only a fraction of the human glycan repertoire. Humans express hundreds of additional cell-surface glycan structures that vary by tissue type, developmental stage, and genetic background, many of which have never been evaluated for clinical immunogenicity because the tools to systematically probe glycan-antibody interactions have only recently become available.

1.2. The xenotransplant perspective

Xenotransplantation provides a compelling demonstration that glycoantigen incompatibility can be a decisive determinant of graft fate. Pigs express several cell-surface glycan structures against which humans can mount antibody responses capable of driving complement-mediated rejection [11,12] — most notably the α-Gal epitope and Neu5Gc, which are absent in all humans, and the Sd^a antigen (*B4GALNT2*), which is absent in some humans who can therefore mount anti-Sd^a responses [10]. Notably, although the vast majority of humans express Sd^a, most still produce antibodies against porcine B4GALNT2 products, suggesting structural differences between the pig and human forms of this glycan. The related Cad antigen ("super Sd^a") shares the same terminal trisaccharide as Sd^a but differs in sub-terminal structure, and the precise relationships among porcine Sd^a, human Sd^a, and human Cad remain incompletely characterized [13] — a case where incomplete glycan structural data and inconsistent nomenclature across disciplines obscure the true nature of a glycoantigen incompatibility. The response to these incompatibilities has driven genetic modification of donor pigs to remove these glycan epitopes [11], demonstrating that identifying and addressing glycan incompatibilities between donor and recipient is a tractable clinical strategy.

2. Glycoantigen challenges

2.1. Complex biosynthesis

Glycans are built by networks of enzymes acting in sequence, making genotype-to-phenotype mapping difficult. Blood group experts use phenotypic designations based on reagent antibody and lectin reactivities that can obscure the underlying carbohydrate chemistry, even though a single antigen phenotype can have multiple structural variants [1].

2.2. Siloed expertise

The different disciplines often speak "different languages." The glycobiology literature encompasses a vast range of glycan biology, making it difficult for clinicians in transfusion and transplant medicine to identify which findings are directly relevant to compatibility; a focused glycoantigen framework could help bridge this gap. Nomenclature differences compound the problem: the same structure may be described in formal glycan notation by a glycobiologist and simply called "α-Gal" by a xenotransplant immunologist. Even within clinical medicine, transfusion specialists refer to the A_2 subtype while transplant programs historically call it "non-A_1" even though *ABO* genotyping has shown that almost all non-reactive samples are in fact *A2* [4]. Meanwhile, when "non-HLA antibodies" are identified in transplant recipients, [14] the possibility that some may be directed against glycoantigens is rarely considered and not readily testable without widespread use of glycan Luminex bead assays [5].

2.3. Incomplete DNA sequence databases

We co-developed the current ISBT Blood Group Database and website [15], which is an essential resource for cataloging blood group alleles and serologic antigen phenotypes. However, it has limitations for the kind of integrated glycoantigen work proposed here. ISBT allele entries are primarily based on coding sequences, and while some *ABO* intronic changes are recorded, these are limited to a few rare intronic changes that alter ABO phenotypes generally due to altered mRNA splicing. Additionally, there are some rare *ABO* alleles with changes in intron 1 regulatory regions and promoter sequences that will soon be added to the ISBT database. However, in practice, many *ABO* alleles in the ISBT tables were originally characterized by sequencing only exons 6 and 7, with no information about the remaining exons, introns, or promoter regions, and historically little attention has been paid to documenting exactly which regions of a gene were examined when an allele was first described. The situation is similar for *FUT1*, *FUT2*, and *FUT3*, where existing allele definitions are often based on partial gene sequences. This creates a practical problem now that more complete sequencing is possible: variants may appear in regions that were never examined for known alleles, making a common, previously described allele look like it is "novel" simply because changes in previously unsequenced regions were not part of the original allele definition. The HLA field has encountered this same issue since its early years and has undertaken systematic efforts to backfill full-length sequences for alleles originally defined from partial data. The glycoantigen work proposed here could contribute updated, more complete sequences back to the ISBT database, and future ISBT database updates could potentially incorporate full-length gene references generated through this effort.

Beyond sequence completeness, the ISBT database records only broad serologic patterns associated with each allele rather than precise glycan structural data. It also largely treats each gene within its respective blood group system in isolation: for example, the H system includes *FUT1* and *FUT2* but leaves it to the user to recognize that these form the biosynthetic basis for ABO antigens, while the Lewis system table includes only *FUT3* despite Lewis antigen expression depending on *FUT2* status as well. It does not capture how allelic variants across multiple genes might interact to produce a given glycoantigen profile on the cell surface. Finally, the ISBT database is appropriately focused on red blood cell antigens, whereas the glycoantigen database I envision would additionally incorporate granular glycan structural data, glycan biosynthetic pathway maps illustrating cross-system interconnections, anti-glycan antibody specificities, tissue-specific expression patterns, and a glycan-binding protein catalog essential for transplantation applications. There is clear synergy between the two resources: the ISBT database provides the authoritative allele nomenclature and red cell serologic framework, while a dedicated glycoantigen database would extend this foundation into the cross-system, tissue-level, and antibody-specificity dimensions that transplantation demands.

2.4. Implementation barriers

Even with the conceptual and technical groundwork in place, obtaining the full datasets required to validate a tool like the Glycoantigen Prediction Pipeline faces substantial obstacles. Although the reference datasets would be collected in specialized labs, all users would at a minimum require *ABO* and *FUT1–3* genotyping to drive the prediction of glycan distribution and the likelihood of antibody formation. Encouragingly, this genotyping is increasingly practical to obtain: *ABO* and *FUT1–3* targets can be spiked into existing HLA next-generation sequencing assays, and PCR-based versions are already offered alongside HLA typing by some vendors, with nanopore-based assays emerging; this allows glycoantigen genotyping to leverage workflows already established in transplantation laboratories. Similarly, standardized glycoantigen nomenclature and reporting formats analogous to those developed for HLA will be needed before results can be reliably exchanged across centers. Finally, it will be necessary to demonstrate clinical utility through multi-center outcome studies to secure regulatory approval and payer coverage. These barriers do not undermine the case for glycoantigen-informed compatibility testing; they define the

practical roadmap by which the field can move from concept to clinical adoption, and the tiered framework proposed in this paper is intended to accommodate that gradual transition.

3. Modern tools enable systematic study

3.1. Sequencing of glycosyltransferase genes

NGS platforms developed for HLA typing have already been extended to include *ABO* genotyping, and are actively being extended to additional glycosyltransferase genes including *FUT1*, *FUT2*, and *FUT3*. Short-read NGS enables high-throughput glycoantigen genotyping at scale, and several groups, including ours, are already gathering this data for population-level characterization of glycosyltransferase allele diversity.

Long-read nanopore sequencing offers a complementary capability, producing full-length gene reference sequences for these loci [16] using workflows analogous to those already established for HLA nanopore assays. These full-length sequences are essential for addressing the reference database gaps described above, and when integrated with the glycan structural and antibody data described below, would provide the genomic foundation for a comprehensive glycoantigen database.

Beyond these currently targeted loci, custom hybrid capture panels could be designed to encompass a broader set of glycosyltransferase genes implicated in blood group and tissue antigen biosynthesis. Nanopore adaptive sampling offers a particularly flexible alternative, selectively sequencing software-specified target regions from a whole-genome library and allowing rapid retargeting to additional loci without redesigning hybridization panels. Whole-exome sequencing (WES) or whole-genome sequencing (WGS) could further expand this scope and help facilitate the identification of novel glycosyltransferase loci not yet associated with clinically relevant glycoantigens. As sequencing costs decline, these broader approaches become increasingly feasible for population-scale glycoantigen characterization.

3.2. Glycan microarrays

By displaying hundreds of defined glycan structures on a single chip [17,18], glycan microarrays allow high-throughput screening of patient sera against a comprehensive panel of glycan structures. This approach has already revealed that pediatric heart transplant recipients develop glycan-subtype-specific tolerance, fine distinctions undetectable by conventional serology [6]. Scaled to clinical cohorts, arrays could systematically map which glycoantigens provoke antibody responses in sensitized transplant candidates, establishing reference datasets that define the glycoantigens of greatest clinical relevance.

3.3. Multiplexed bead immunoassays

Once array-based discovery has identified the glycoantigens of greatest clinical relevance, those targets can be conjugated onto Luminex beads to create focused, clinically deployable assays [19]. Luminex-based bead assays can quantify antibodies against specific glycan subtypes in a format already familiar to histocompatibility laboratories, and because this platform is the same one used for routine anti-HLA antibody detection, glycoantigen-specific antibody testing could be integrated into existing clinical workflows without new instrumentation [5].

3.4. High-resolution mass spectrometry

Glycoproteomics approaches can characterize the glycan structures present on specific cell types and tissues with a level of structural detail that complements array- and bead-based methods. Applied to transplant-relevant tissues such as vascular endothelium and kidney tubular epithelium, mass spectrometry provides direct structural confirmation of the glycoantigen landscape a recipient's immune system will encounter, going beyond what can be inferred from genomic data alone.

3.5. Transcriptomic profiling

Bulk RNA-seq and single-cell RNA-seq (scRNA-seq) can map glycosyltransferase gene expression across tissues and cell types, revealing which biosynthetic enzymes are actively transcribed in transplant-relevant organs and thereby predicting which glycoantigens are likely to be presented on the cell surface. This information speaks directly to the central clinical question in solid organ transplantation: for a given donor organ, which glycans are being actively produced, and could any of those glycans trigger an immune response in a recipient whose own tissues do not express them? Substantial expression data already exist in publicly accessible resources that could be curated and cross-referenced with the proposed glycoantigen database. However, the greatest value will come from multi-modal samples in which gene expression, glycan structural characterization, and antibody reactivity data are all generated from the same specimens, because it is only through such integrated measurements that we can determine how transcriptomic output translates to actual glycoantigen surface display and, ultimately, to immunological relevance in the transplant setting. Resolving glycoantigen expression to the cell-type level within an organ is more data-intensive than organ-level phenotyping but is increasingly tractable with single-cell transcriptomics and glycan-specific reagents including lamprey-derived antibodies and monoclonal antibodies applied by immunohistochemistry [20].

4. Future directions

Establishing glycoantigens as a recognized field will require coordinated efforts across multiple fronts:

4.1. Near-term priorities: building the foundation

The most critical first step is creating a comprehensive integrated glycoantigen database linking genomic data (glycosyltransferase full gene sequences and allele frequencies), glycan structural data (mass spectrometry profiles, biosynthetic pathways, and glycan array binding patterns), tissue-specific expression data, and serologic data (traditional typing, antibody specificities, and lectin binding profiles). This database should connect to existing glycan databases, genomic databases, and the ISBT Blood Group Database [15] (Fig. 1).

Database development should follow a phased approach, beginning with schema design validated against mock data, then populating the database with well-characterized reference samples before expanding to rare samples and unusual phenotypes. Work on this initial phase has already begun, in collaboration with experts in blood group genetics, transplant compatibility testing, glycomics, and antigen immunology. Critically, because the database links genomic, structural, and serologic data from the same characterized samples, it can serve as more than a static repository. A user could input a glycosyltransferase gene sequence, match it to a known allele, and retrieve the predicted enzyme activity. The database could then combine predicted enzyme activities across multiple glycosyltransferase loci to infer, based on mass spectrometry data from previously characterized samples with similar genotypes, which glycoantigens are likely present or absent on specific tissue types. For each predicted missing glycoantigen, the database could further report the likelihood that an individual will form antibodies against that structure, drawing on glycan array and Luminex bead immunoassay data from characterized cohorts. I term this the Glycoantigen Prediction Pipeline (Fig. 2): a complete genotype-to-enzyme-to-glycan-to-antibody prediction workflow that represents the core translational value of an integrated, multi-modal sample database. The database should also include a glycan-binding protein (GBP) catalog documenting monoclonal antibodies, lamprey-derived antibodies [20],

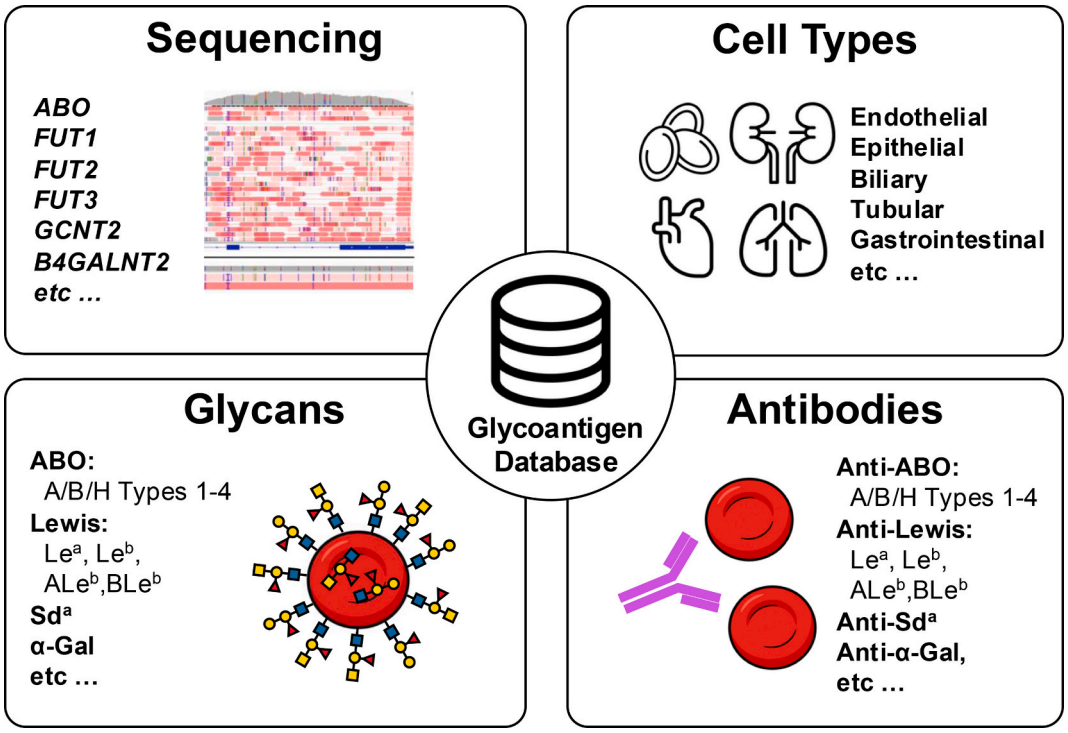


Fig. 1. Integrated Glycoantigen Database Architecture. The proposed glycoantigen database serves as a central resource linking four interconnected data domains: (1) Sequencing, genomic characterization of glycosyltransferase genes (e.g., *ABO*, *FUT1*, *FUT2*, *FUT3*, *GCNT2*, *B4GALNT2*) including allele variants and population frequencies; (2) Cell Types, tissue-specific glycoantigen expression patterns across organs relevant to transplantation (including endothelial, epithelial, biliary, tubular, and gastrointestinal cell types); (3) Glycans, structural characterization of cell-surface glycan repertoires spanning the ABH, Lewis, Sda, and α-Gal systems; and (4) Antibodies, serologic reactivity profiles including anti-glycan antibody specificities (anti-ABH, anti-Lewis, anti-Sda, and anti-α-Gal). Integration of these domains enables comprehensive genotype-to-phenotype mapping and identification of clinically relevant glycoantigen mismatches.

lectins, galectins, siglecs, and selectins with their glycan target specificities and validated applications (e.g., flow cytometry, immunohistochemistry, Luminex), providing investigators with a practical resource for selecting appropriate tools to study or test for specific glycoantigens.

The 19th International HLA & Immunogenetics Workshop (IHIW 19) provides an established collaborative framework for advancing this effort. The IHIW ABO workshop component will collect next-generation sequencing (NGS) short-read data for *ABO* (exons 2–7) and *FUT1*, *FUT2*, and *FUT3* (core exons), along with Luminex bead-based antibody data. Selected samples from this workshop will additionally undergo nanopore long-read sequencing to generate phased, full-gene-length haplotype sequences where needed, thereby establishing a set of validated full-gene reference sequences. Beyond this initial workshop endeavor, an international glycoantigen consortium should be formed, bringing together stakeholders from glycobiology, bioinformatics, transplant immunology, and transfusion medicine to govern database development and ensure data quality and standardization.

4.2. *Medium-term goals: translation to discovery*

With the foundational database established, the next phase involves launching systematic discovery studies using glycan arrays to screen sera from patients with unexplained antibody-mediated rejection, cross-referencing antibody specificities with donor glycoantigen profiles. Glycan mass spectrometry will be particularly valuable at this stage for characterizing the actual glycan structures present on cells and tissues, moving beyond genomic inference to direct structural confirmation.

A practical challenge is that while archived DNA and sera are readily available, red blood cells are not typically archived; most genomic studies lyse red cells during DNA extraction. However, RBCs can be frozen in Glycerolyte 57 and stored in standard −80 °C freezers [21], and I propose that this approach should be adopted routinely to build paired genomic-glycomic datasets from characterized samples. Mass spectrometry should also be performed on selected fresh samples characterized by NGS, and excess clinical biopsy samples and unallocated solid organs from deceased donors could serve as valuable sources for tissue-specific glycan characterization.

Prospective observational studies correlating donor-recipient glycoantigen mismatches with transplant outcomes should be conducted across multiple centers to establish clinical relevance. In parallel, therapeutic strategies including enzymatic antigen removal [22], glycan-targeted immunosuppression, and glycoengineered cell therapies should be explored, alongside the development of xeno-specific glycoantigen Luminex beads for clinical monitoring of antibody responses directed against non-human glycan epitopes.

4.3. *Long-term vision: clinical integration*

The ultimate goal is to integrate glycoantigen profiling into standard compatibility testing for solid organ transplantation, analogous to current HLA typing. This would enable “glycoantigen-matching” algorithms that optimize donor-recipient pairing based on comprehensive glycan compatibility, and facilitate clinical trials of glycoantigen-targeted interventions with regulatory approval pathways.

Although the initial focus of this work is transfusion and transplantation, anti-glycan antibodies play clinically significant roles across medicine [23]. Alpha-gal syndrome, in which IgE against the same α-Gal epitope that is a key target in xenotransplantation causes delayed allergic reactions to mammalian meat, illustrates how a single glycoantigen can have consequences spanning transplant immunology and allergy. Anti-ganglioside antibodies (e.g., anti-GQ1b, anti-GM1) serve as established diagnostic markers in Guillain-Barré spectrum disorders, anti-I antibodies drive cold agglutinin disease and autoimmune hemolytic anemia, anti-mannan antibodies (ASCA) are used in inflammatory

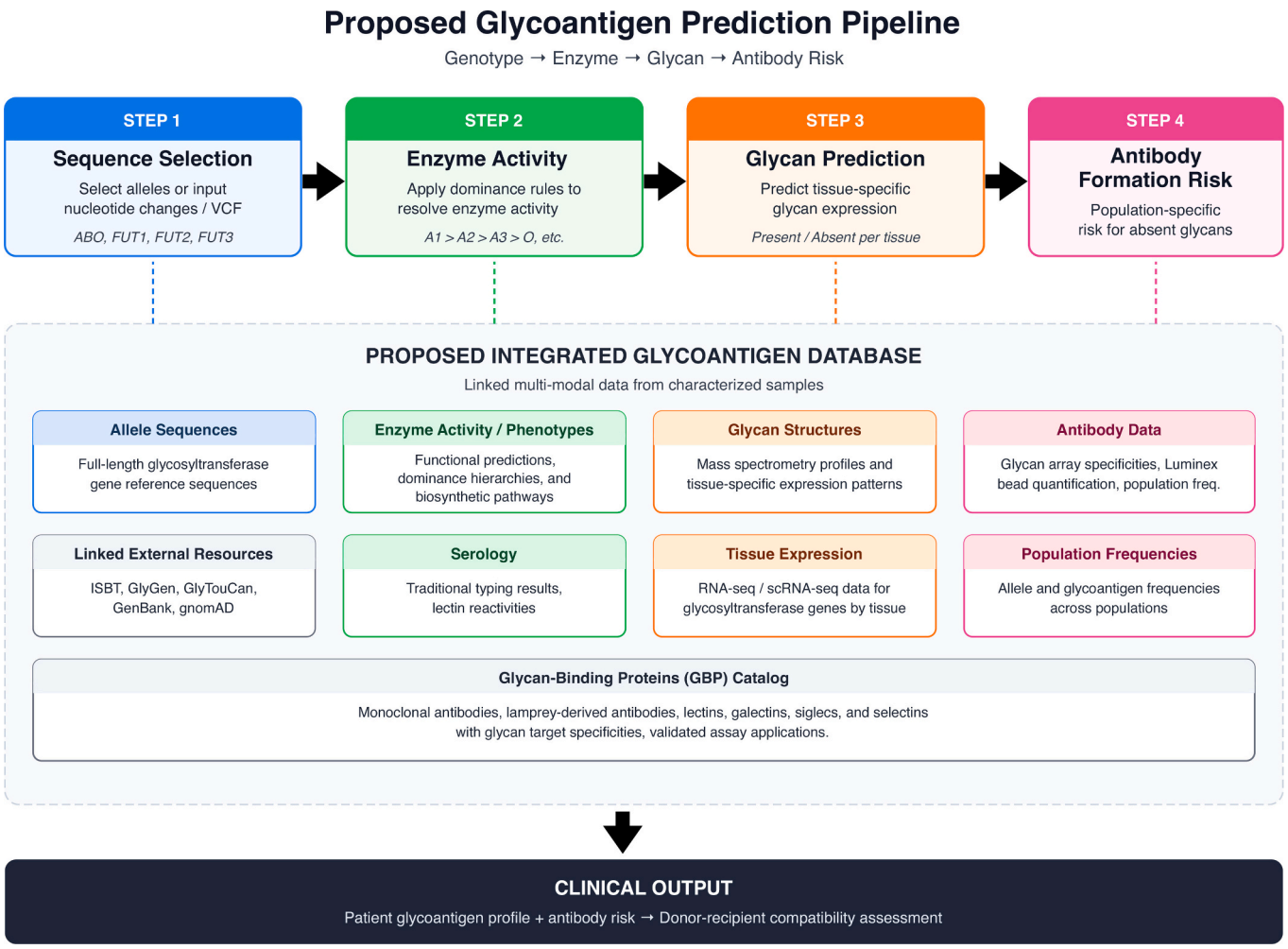


Fig. 2. Proposed Glycoantigen Prediction Pipeline. The pipeline illustrates the genotype-to-enzyme-to-glycan-to-antibody prediction workflow. Step 1: a user inputs glycosyltransferase gene sequences (via allele selection, nucleotide changes, or VCF file) and matches them to known alleles. Step 2: enzyme activities are resolved by applying dominance rules across allele pairs. Step 3: predicted enzyme activities are combined across multiple loci to infer tissue-specific glycoantigen profiles (present or absent), drawing on mass spectrometry and expression data from characterized samples with similar genotypes. Step 4: for each absent glycoantigen, the pipeline estimates the likelihood of antibody formation based on glycan array and Luminex bead immunoassay data from characterized cohorts, stratified by population. The pipeline is supported by the proposed integrated glycoantigen database (bottom), which encompasses eight linked modules: allele sequences, enzyme activity and phenotype predictions, glycan structures, antibody data, linked external resources (ISBT, GlyGen, GlyTouCan, GenBank, gnomAD), serology, tissue expression, and population frequencies. Color coding indicates the primary data source for each pipeline stage. A glycan-binding protein (GBP) catalog documents monoclonal antibodies, lamprey-derived antibodies, lectins, galectins, siglecs, and selectins with their glycan target specificities and validated assay applications. All steps converge on a clinical output: a patient-specific glycoantigen profile with antibody risk assessment for donor-recipient compatibility evaluation.

bowel disease serologic panels, anti-capsular polysaccharide antibodies are foundational to vaccine response assessment and immunodeficiency evaluation, and carbohydrate tumor antigens such as CA19–9 (sialyl-Le^a) are used in oncology monitoring. A comprehensive glycoantigen database could eventually incorporate annotations linking glycan structures to these broader clinical contexts, creating a resource that extends well beyond its transfusion and transplant origins.

5. Conclusion

Glycoantigens represent a promising frontier at the intersection of glycobiology, transfusion medicine, and transplant immunology. The ABO blood group system taught medicine one of its earliest and most important lessons in compatibility, yet only now are the tools available to appreciate the full complexity of glycoantigens and their relevance beyond red blood cell transfusion. By recognizing glycoantigens as a unified concept, connecting glycosyltransferase-mediated biosynthesis to the clinical reality that a single glycan motif can determine transfusion and graft fate, this approach establishes the framework for investigations that could meaningfully improve patient outcomes. The database, Glycoantigen Prediction Pipeline, and consortium proposed here provide a concrete roadmap for ensuring that important compatibility factors are systematically examined, with the potential to extend into other areas of clinical medicine where anti-glycan antibodies play significant roles.

CRediT authorship contribution statement

William J. Lane: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author used Claude (Anthropic, Claude Opus 4.6) and ChatGPT (OpenAI, ChatGPT 5.2 Thinking) to assist with editing for readability and language and with figure generation. The author reviewed and edited all output and takes

full responsibility for the content of this publication.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: William Lane reports a relationship with CareDx Inc. that includes: consulting or advisory. William Lane reports a relationship with International Society of Blood Transfusion that includes: consulting or advisory. William Lane has patent with royalties paid to Thermo Fisher Scientific. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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